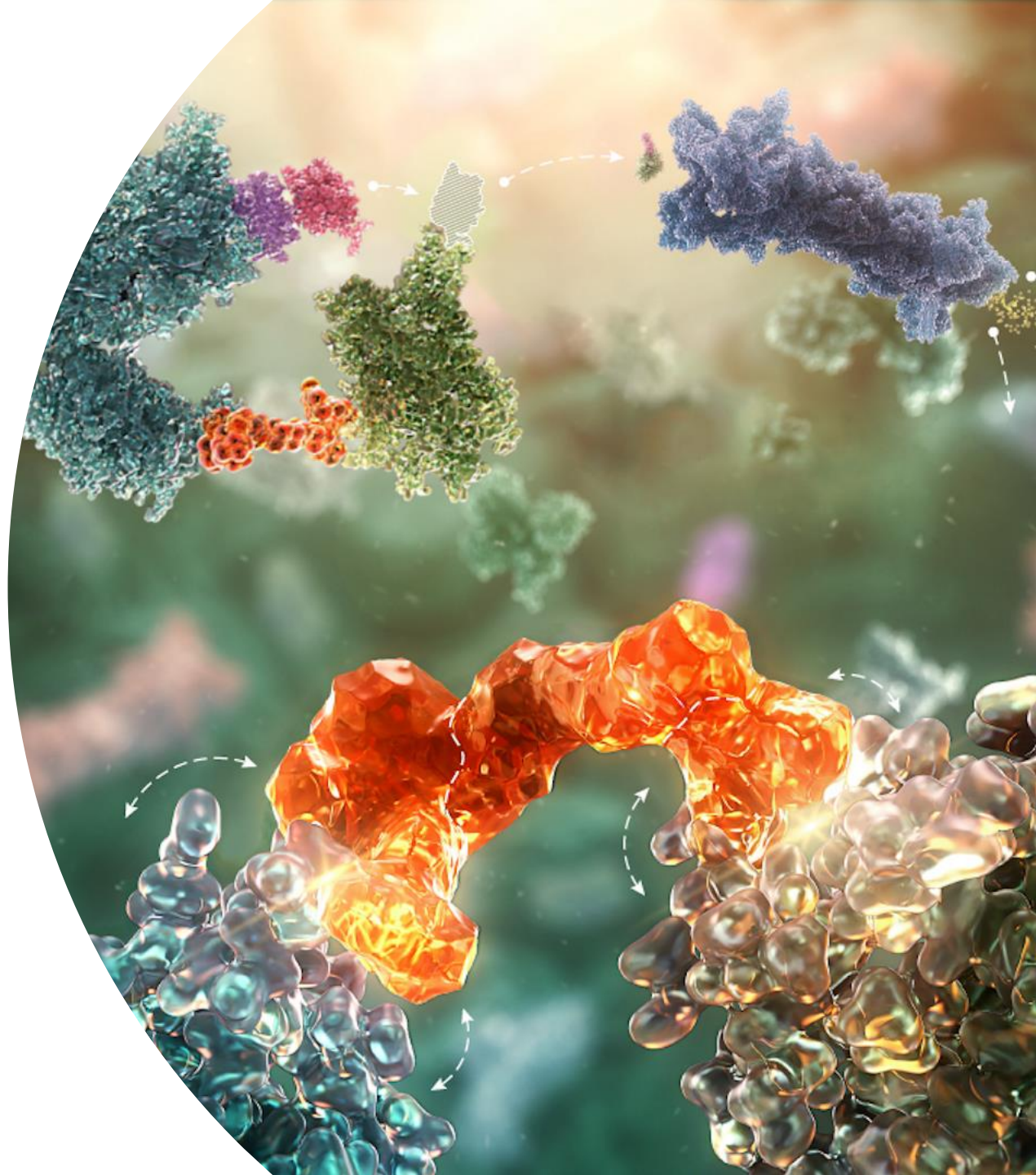


# Effect of Inner Diameter, Filter Length, and Pore Size on Hollow Fiber Filter Fouling in Perfusion Cell Culture

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11Oct24



# Agenda

- 1 Tangential flow filtration (TFF) introduction
- 2 High filter flux screen 1: probing effect of length and lumen ID
- 3 High filter flux screen 2: probing effect of pore size
- 4 Reduced flux confirmation study
- 5 Conclusions

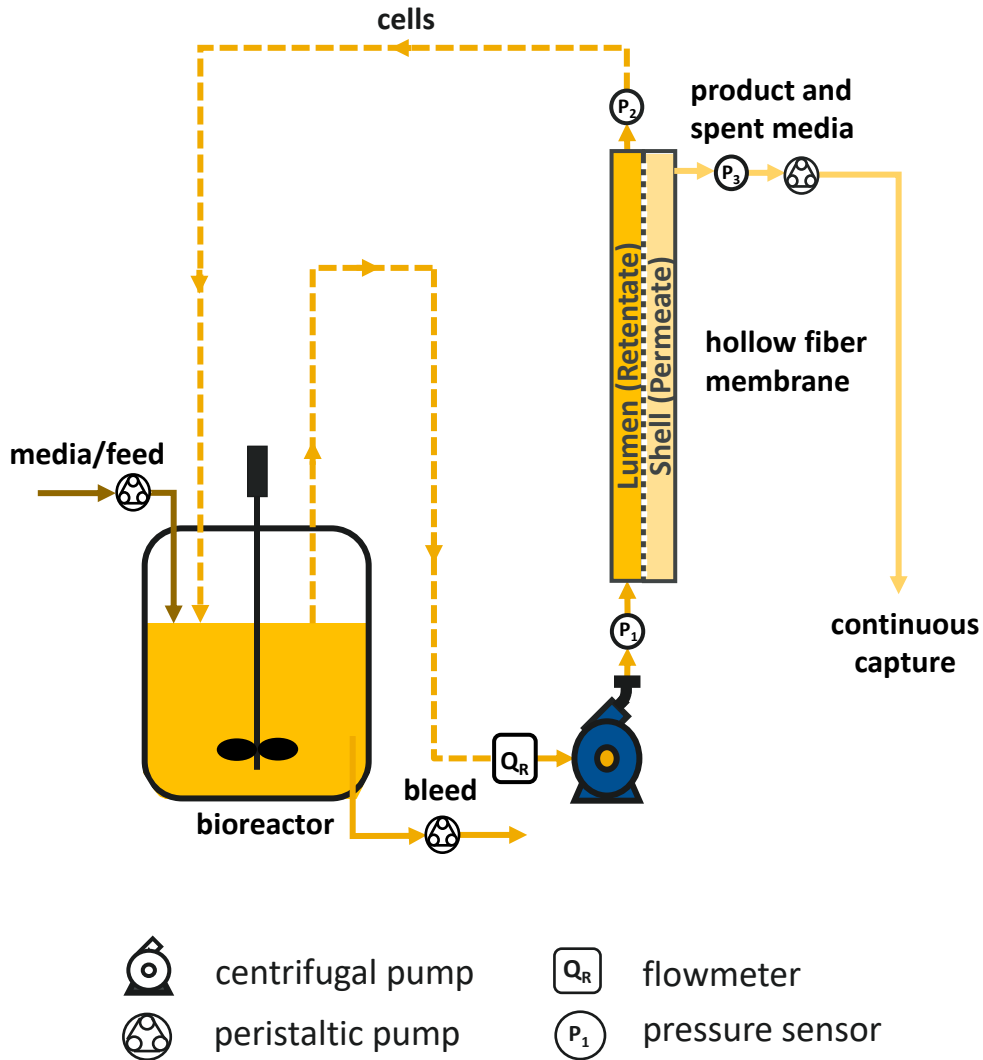


1

# Tangential Flow Filtration (TFF)



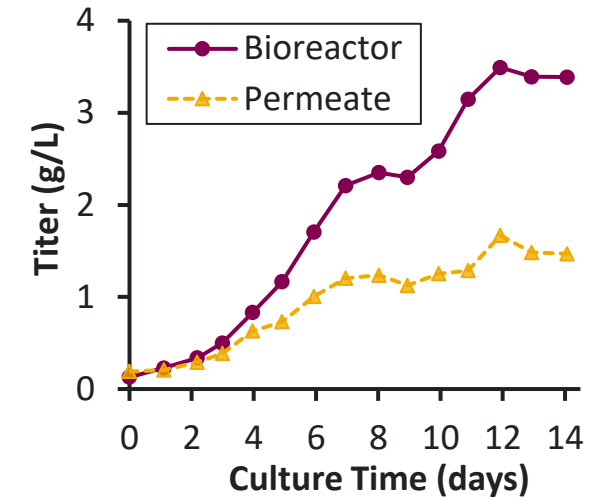
# TFF perfusion cell culture



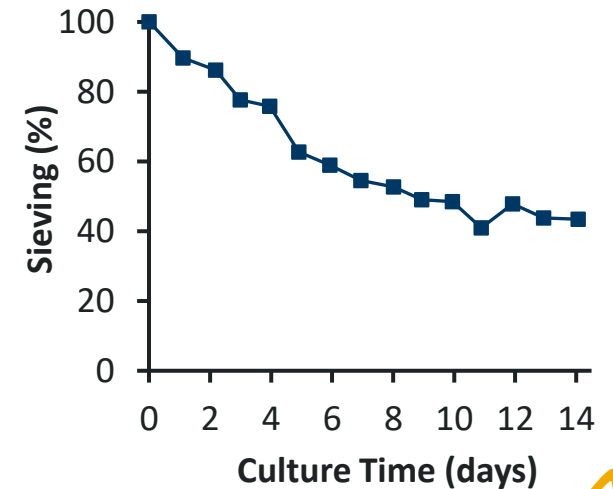
- Higher cell densities and volumetric productivities ( $\text{g/L}_{\text{brx}}$ ) achieved over traditional fed-batch processes
- Continuous harvest allows for significant downstream advantages

| TFF Advantages                                                                                                                                                        | TFF Disadvantages                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Scalability compared to ATF process (&gt;2000L)</li> <li>• Compatibility with different hollow fiber manufactures</li> </ul> | <ul style="list-style-type: none"> <li>• Higher shear due to use of centrifugal pump</li> <li>• <b>Membrane fouling affecting product sieving</b></li> </ul> |

## Filter fouling



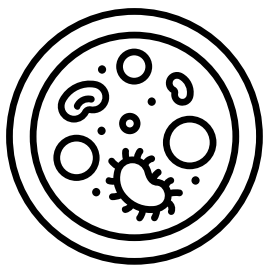
$$\text{Sieving (\%)} = \frac{C_{\text{perm}}}{C_{\text{brx}}} \times 100$$



# Managing filter fouling in perfusion cell culture

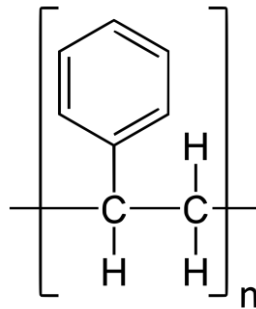
## Cell Health

- Nutrient Feed
- Perfusion Rate
- Mechanical Stress



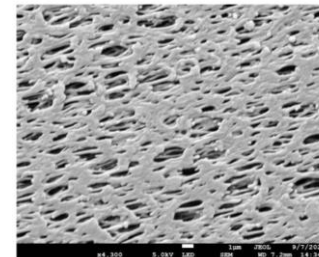
## Membrane Chemistry

- PS
- PES
- mPES
- PVDF
- ME



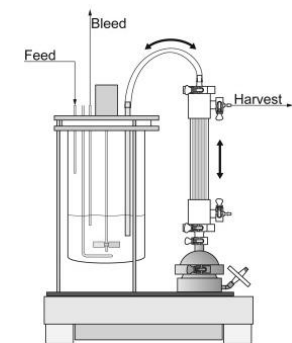
## Fiber Geometries

- Fiber length
- Lumen diameter
- Pore size



## Operating Conditions

- ATF
- TFF
- Stacked TFF
- LowFlowTFF
- HPTFF



Pinto, N. D. S.; Brower, 2020. <https://doi.org/10.1002/bit.27504>.

Sharma et al., 2021. <https://doi.org/10.3390/membranes11020120>.

Karst et al., 2016. <https://doi.org/10.1016/j.bej.2016.02.003>.



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High flux screen 1:  
probing effect of lumen  
ID and fiber length



# Lumen ID and fiber length high flux screen: experimental design

Due to differences in the geometry of the hollow fibers, parameters were chosen to match shear rate, flux, and exchange rate

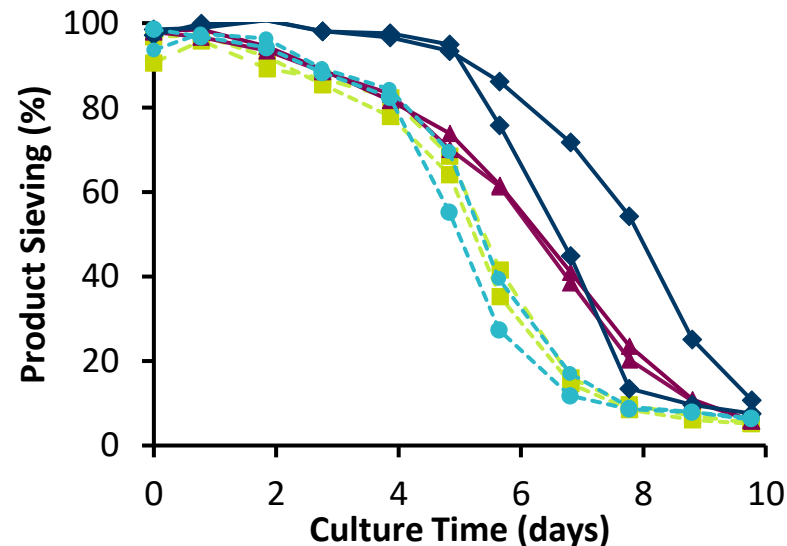
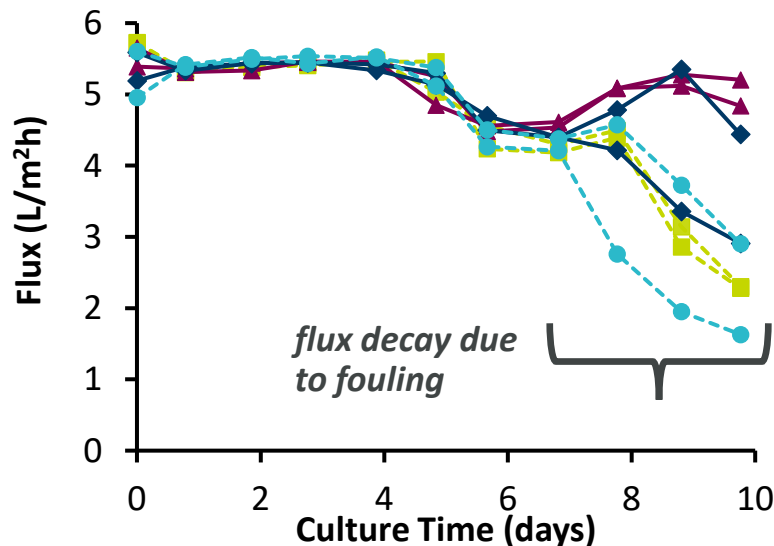
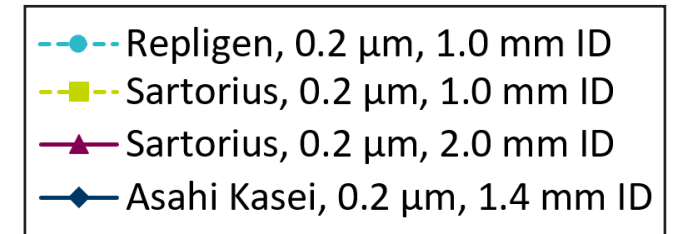
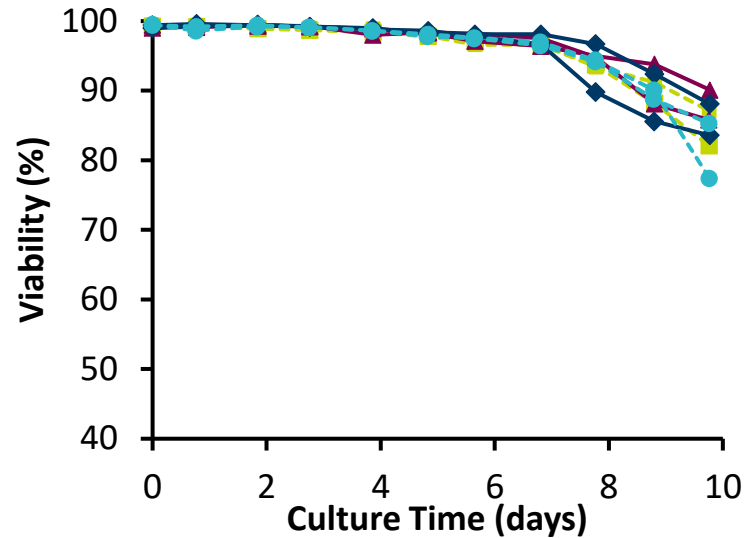
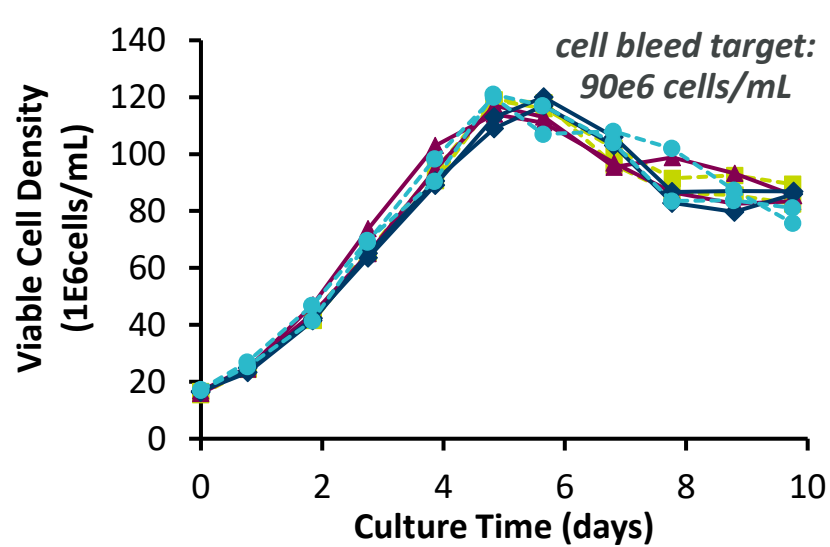
|                         |     |
|-------------------------|-----|
| Shear Rate ( $s^{-1}$ ) | 600 |
| Flux (LMH)              | 5.5 |
| Exchange Rate (VVD)     | 1.2 |
| Scale (L)               | 2   |

| Manufacturer | Part Number   | Pore Size ( $\mu m$ ) | Chemistry | ID (mm) | Area ( $cm^2$ ) | Length (cm) | Fiber count |
|--------------|---------------|-----------------------|-----------|---------|-----------------|-------------|-------------|
| Repligen     | T04-P20U-10-N | 0.2                   | PES       | 1.0     | 180             | 41.5        | 14          |
| Sartorius    | AU92010EXP12  | 0.2                   | mPES      | 1.0     | 155             | 27.3        | 18          |
| Sartorius    | AU92020EXP12  | 0.2                   | mPES      | 2.0     | 155             | 27.3        | 9           |
| Asahi Kasei  | UMP-0047R     | 0.2                   | PVDF      | 1.4     | 200             | 19.8        | 21          |



# Lumen ID and fiber length high flux screen: culture performance

No impact of filter type on cell growth and health; flux target and higher sieving maintained longer with larger lumen ID



# Lumen ID and fiber length high flux screen: hydraulic membrane resistance

Larger lumen ID filters maintained the lowest membrane resistance

Assuming cylindrical pores, the resistance ( $R_t$ ) based on Darcy's equation is:

$$J = \frac{\text{TMP}}{\mu R_t}$$

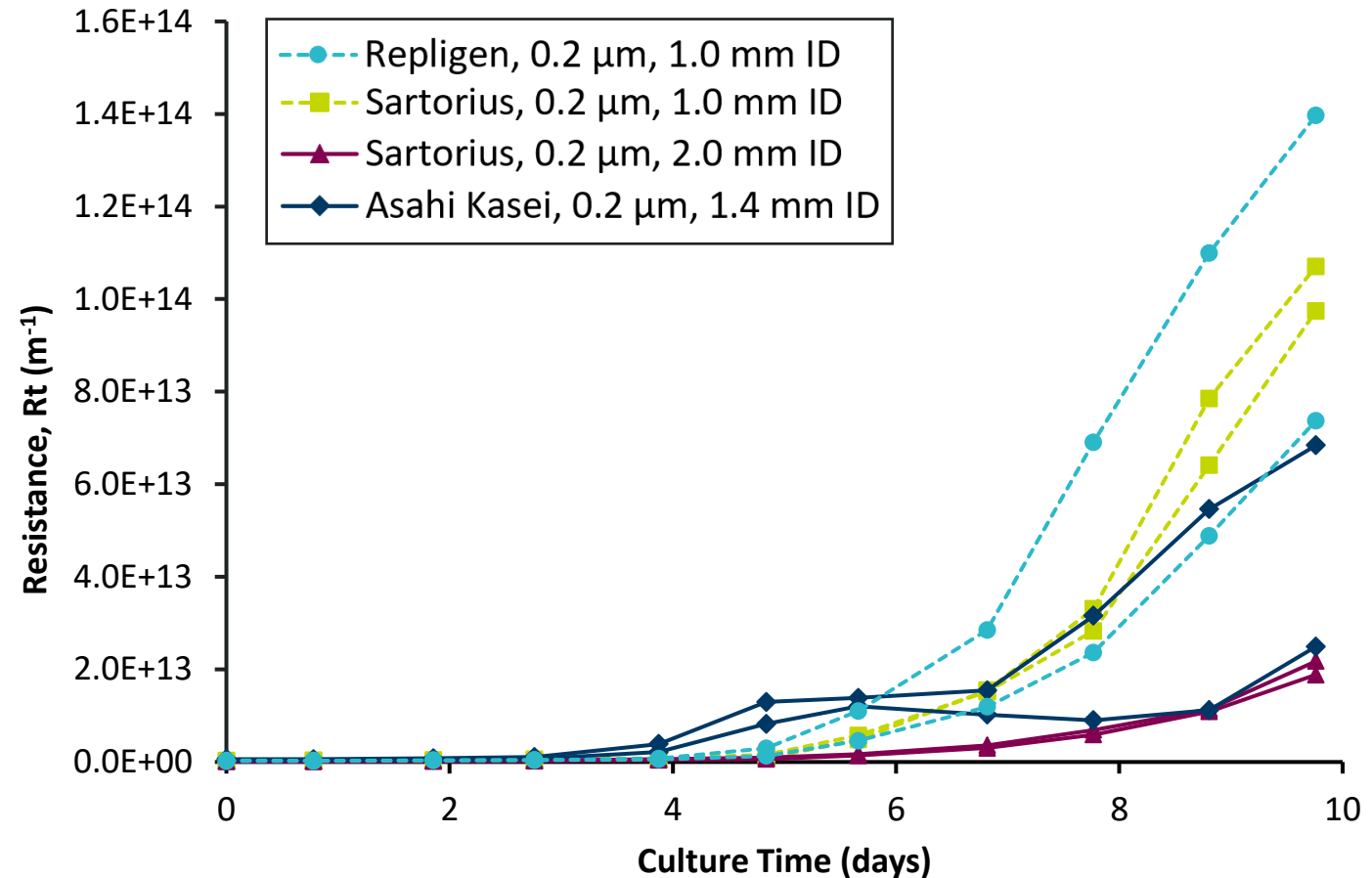
$$R_t = \frac{8\delta_m}{r_p^2 \varepsilon}$$

$\delta_m$  = membrane thickness

$\varepsilon$  = membrane porosity

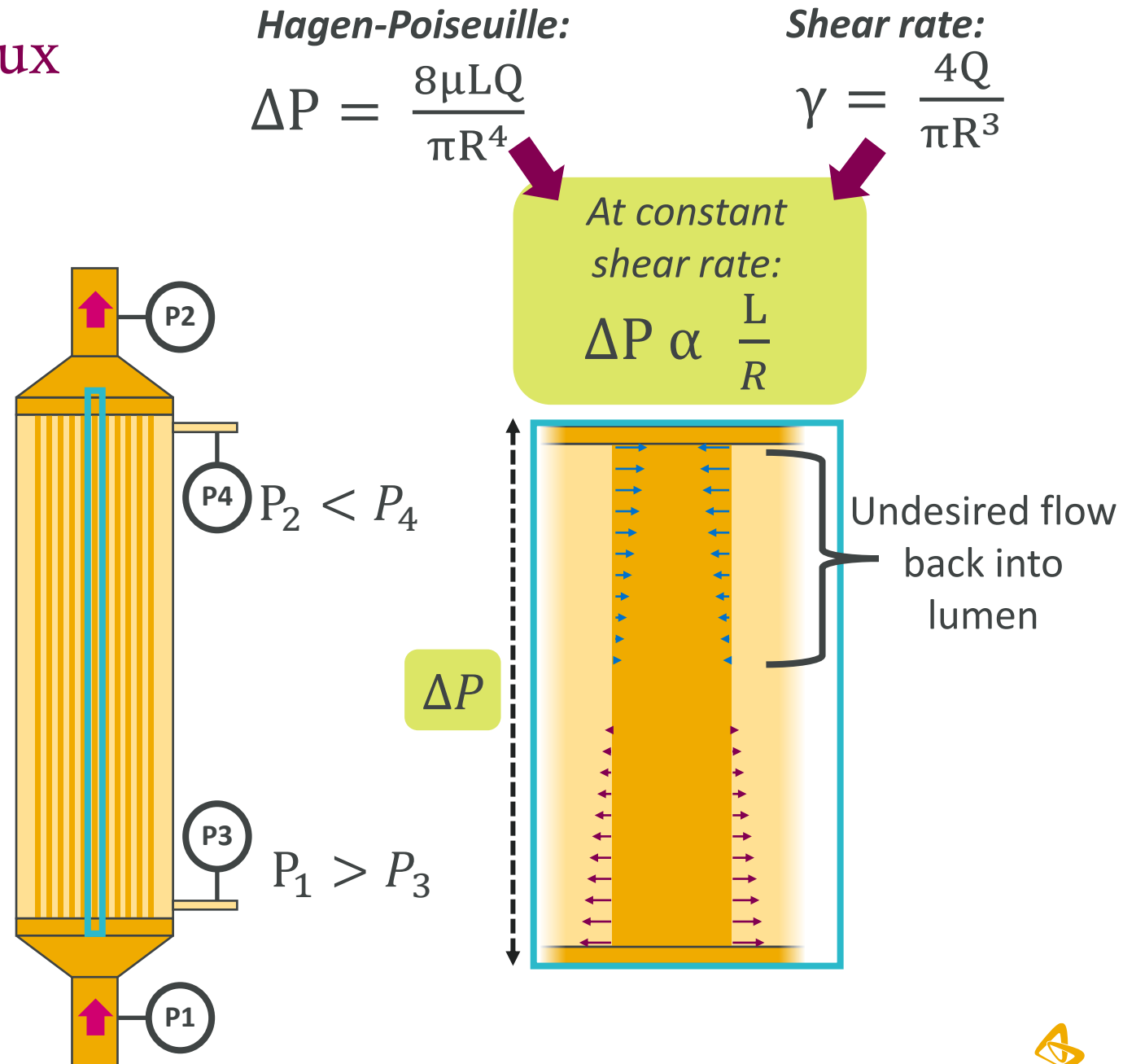
$\mu$  = viscosity

$r_p$  = pore radius



# Lumen ID and fiber length high flux screen: conclusions

- Larger lumen ID hollow fibers showed improved product sieving and lower hydraulic membrane resistance
- Combination of shorter hollow fiber and larger lumen ID reduces axial pressure drop and flow back into the lumen



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High flux screen 2:  
probing effect of pore size



# Pore size high flux screen: experimental design

Due to differences in the geometry of the hollow fibers, parameters were chosen to match shear rate, flux, and exchange rate

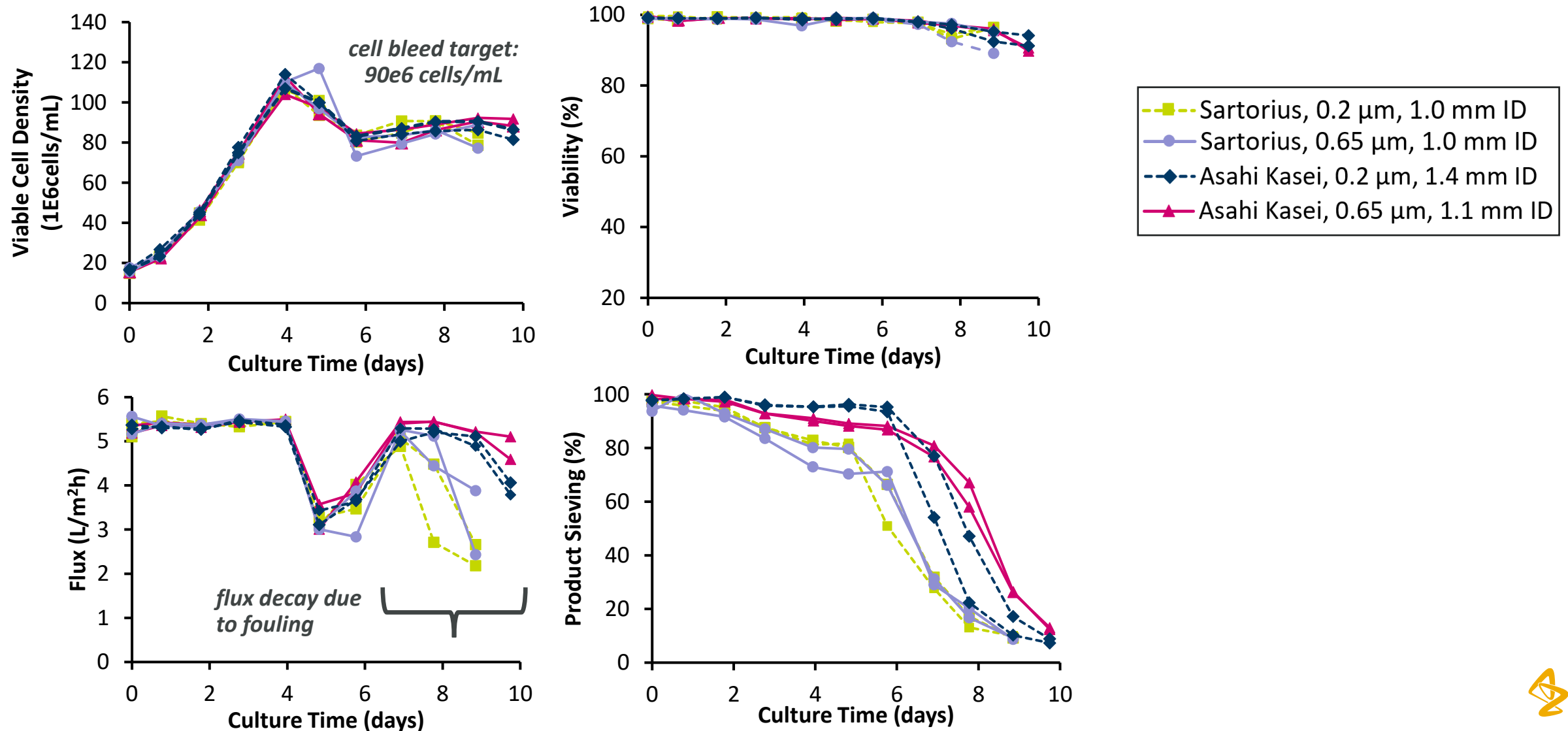
|                                |     |
|--------------------------------|-----|
| Shear Rate ( $\text{s}^{-1}$ ) | 600 |
| Flux (LMH)                     | 5.5 |
| Exchange Rate (VVD)            | 1.2 |
| Scale (L)                      | 2   |

| Manufacturer | Part Number  | Pore Size ( $\mu\text{m}$ ) | Chemistry | ID (mm) | Area ( $\text{cm}^2$ ) | Length (cm) | Fiber count |
|--------------|--------------|-----------------------------|-----------|---------|------------------------|-------------|-------------|
| Sartorius    | AU92010EXP12 | 0.2                         | mPES      | 1.0     | 155                    | 27.3        | 18          |
| Sartorius    | AU96510EXP12 | 0.65                        | mPES      | 1.0     | 155                    | 27.3        | 18          |
| Asahi Kasei  | UMP-0047R    | 0.2                         | PVDF      | 1.4     | 200                    | 19.8        | 21          |
| Asahi Kasei  | UJP-0047R    | 0.65                        | PVDF      | 1.1     | 200                    | 19.8        | 27          |



# Pore size high flux screen: culture performance

No impact of filter type on cell growth and health; larger pore size filters showed no significant improvement in sieving



# Pore size high flux screen: hydraulic membrane resistance

Larger pore sized filters had reduced membrane resistance

Asahi UJP-0047R filter (0.65  $\mu\text{m}$ ) maintained the lowest membrane resistance

Assuming cylindrical pores, the resistance ( $R_t$ ) based on Darcy's equation is:

$$J = \frac{\text{TMP}}{\mu R_t}$$

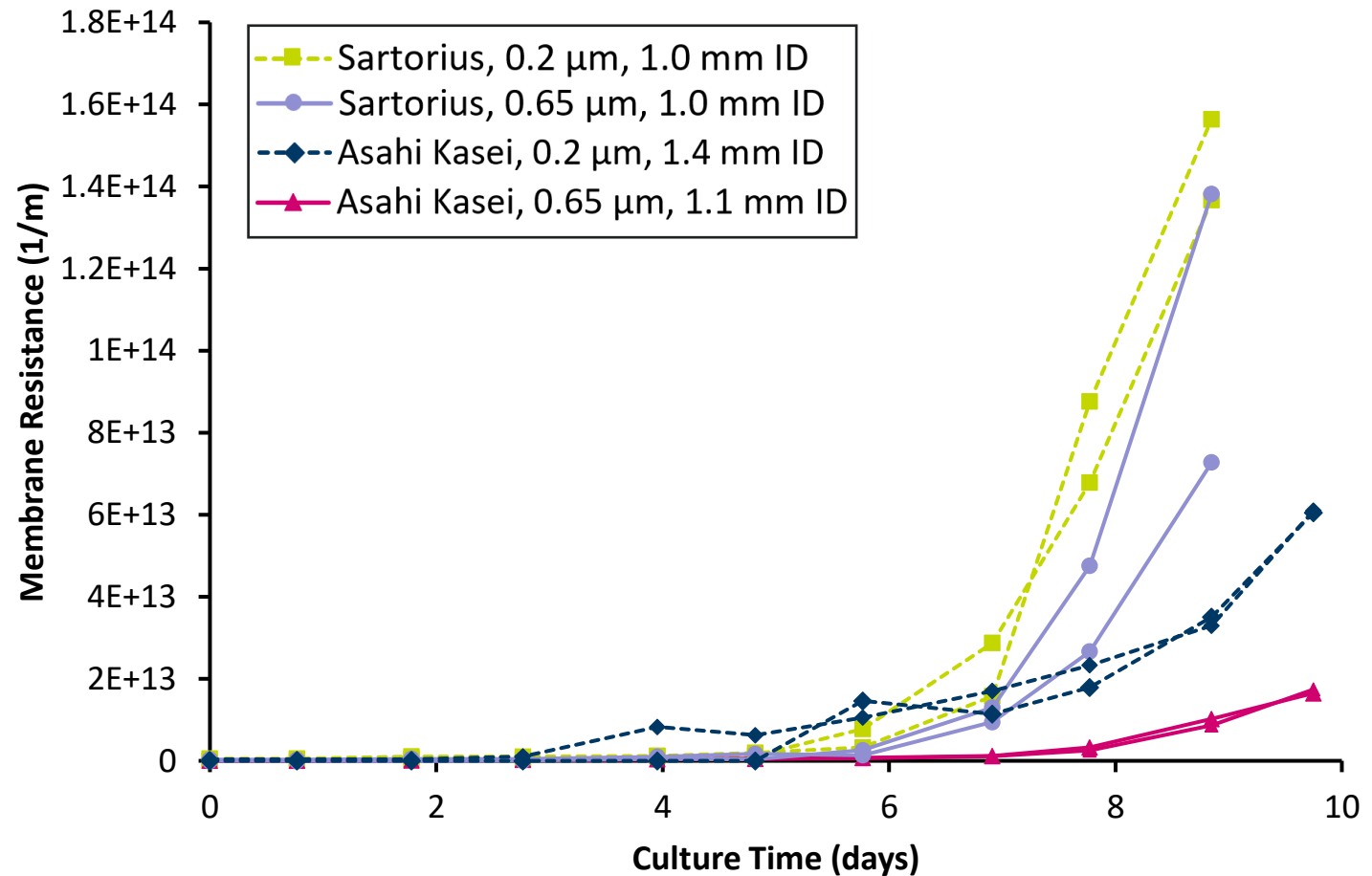
$$R_t = \frac{8\delta_m}{r_p^2 \varepsilon}$$

$\delta_m$  = membrane thickness

$\varepsilon$  = membrane porosity

$\mu$  = viscosity

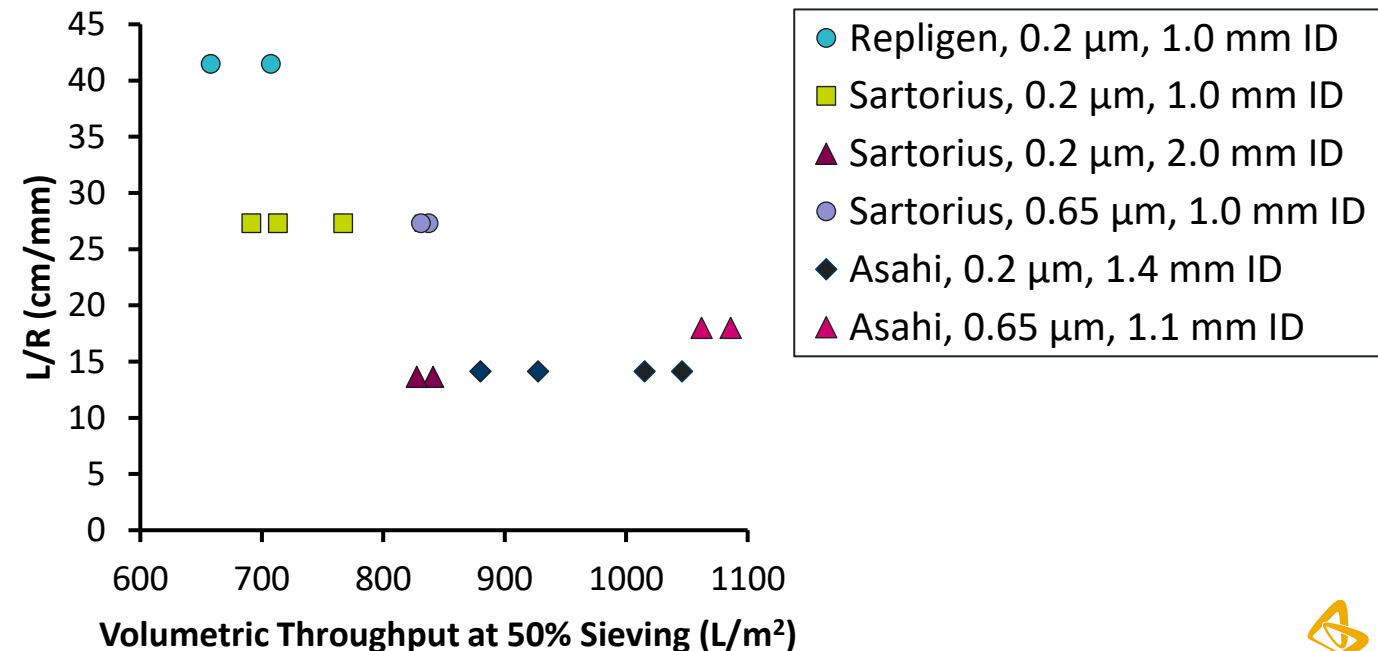
$r_p$  = pore radius



# Pore size high flux screen: conclusions

- Larger pore size filters (0.65  $\mu\text{m}$  vs. 0.2  $\mu\text{m}$ ) reduced hydraulic membrane resistance but did not result in significant improvements in product sieving
- Filters manufactured by Asahi Kasei showed improved performance over Sartorius filters due to a combination of filter geometry/chemistry

| Manufacturer | Lumen ID | Fiber Length | Membrane Chemistry |
|--------------|----------|--------------|--------------------|
| Asahi Kasei  | 1.1-1.4  | 19.8         | PVDF               |
| Sartorius    | 1.0      | 27.3         | mPES               |



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## Reduced flux verification study



# Reduced flux : experimental design

Due to differences in the geometry of the hollow fibers, parameters were chosen to match shear rate, flux, and exchange rate

|                     |      |
|---------------------|------|
| Flux (LMH)          | 1.02 |
| Exchange Rate (VVD) | 1.2  |
| Scale (L)           | 2    |

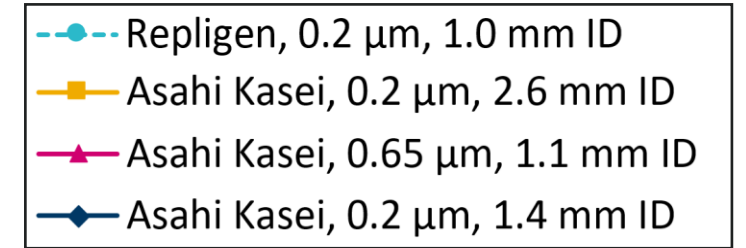
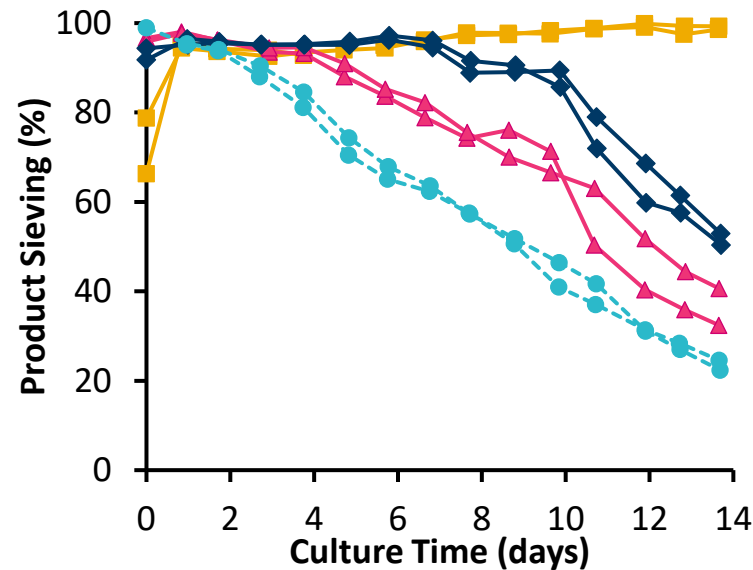
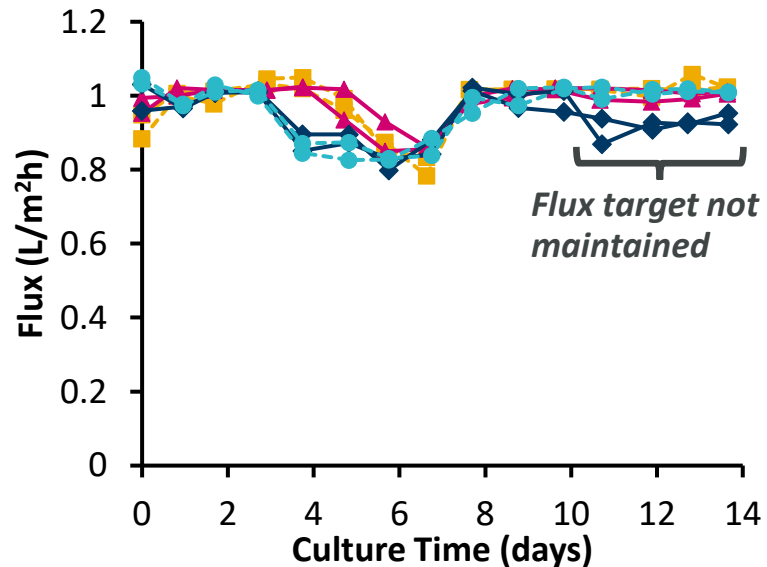
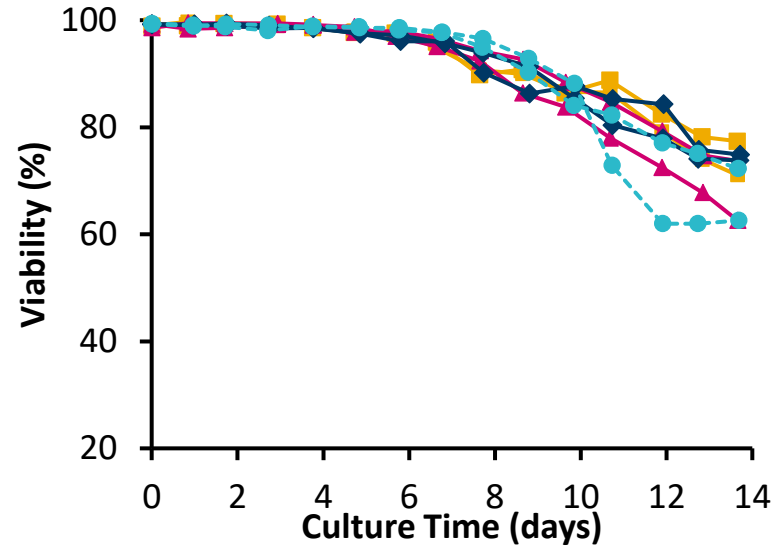
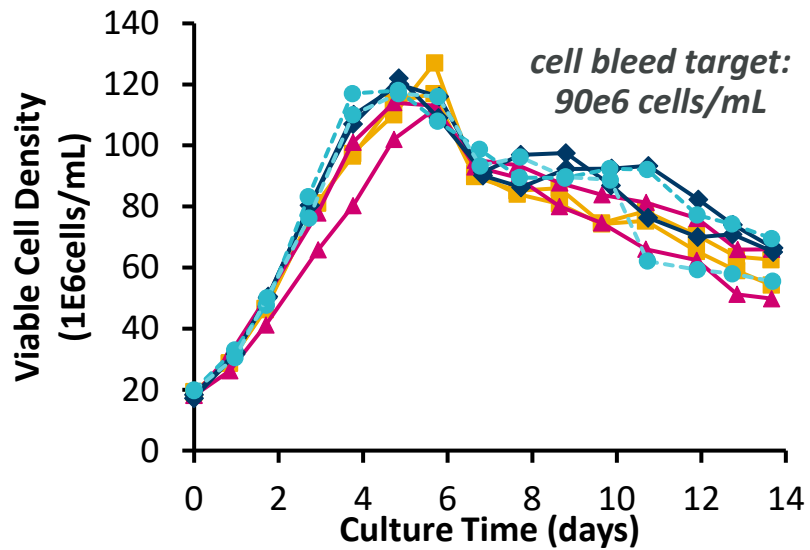
Top performing filters were tested at a flux setpoint that would maintain operable TMPs for a full-length process (14 days)

| Manufacturer | Part Number   | Pore Size (um) | Chemistry | ID (mm) | Area (cm <sup>2</sup> ) | Length (cm) | Fiber Count | Shear Rate (s <sup>-1</sup> ) |
|--------------|---------------|----------------|-----------|---------|-------------------------|-------------|-------------|-------------------------------|
| Repligen     | S04-P20U-10-N | 0.2            | PES       | 1.0     | 980                     | 41.5        | 75          | 600                           |
| Asahi Kasei  | UMP-153       | 0.2            | PVDF      | 2.6     | 800                     | 19.8        | 50          | 300                           |
| Asahi Kasei  | UJP-1047R     | 0.65           | PVDF      | 1.1     | 1000                    | 19.8        | 135         | 600                           |
| Asahi Kasei  | UMP-1047R     | 0.2            | PVDF      | 1.4     | 900                     | 19.8        | 100         | 600                           |



# Reduced flux: cell culture performance

No impact of filter type on cell growth and health; UMP-153 filter (2 mm ID) maintained highest sieving followed by UMP-1047R (1.4 mm ID)



# Reduced flux: hydraulic membrane resistance

UMP-153 filters (2 mm ID) maintained the lowest membrane resistance

Assuming cylindrical pores, the resistance ( $R_t$ ) based on Darcy's equation is:

$$J = \frac{\text{TMP}}{\mu R_t}$$

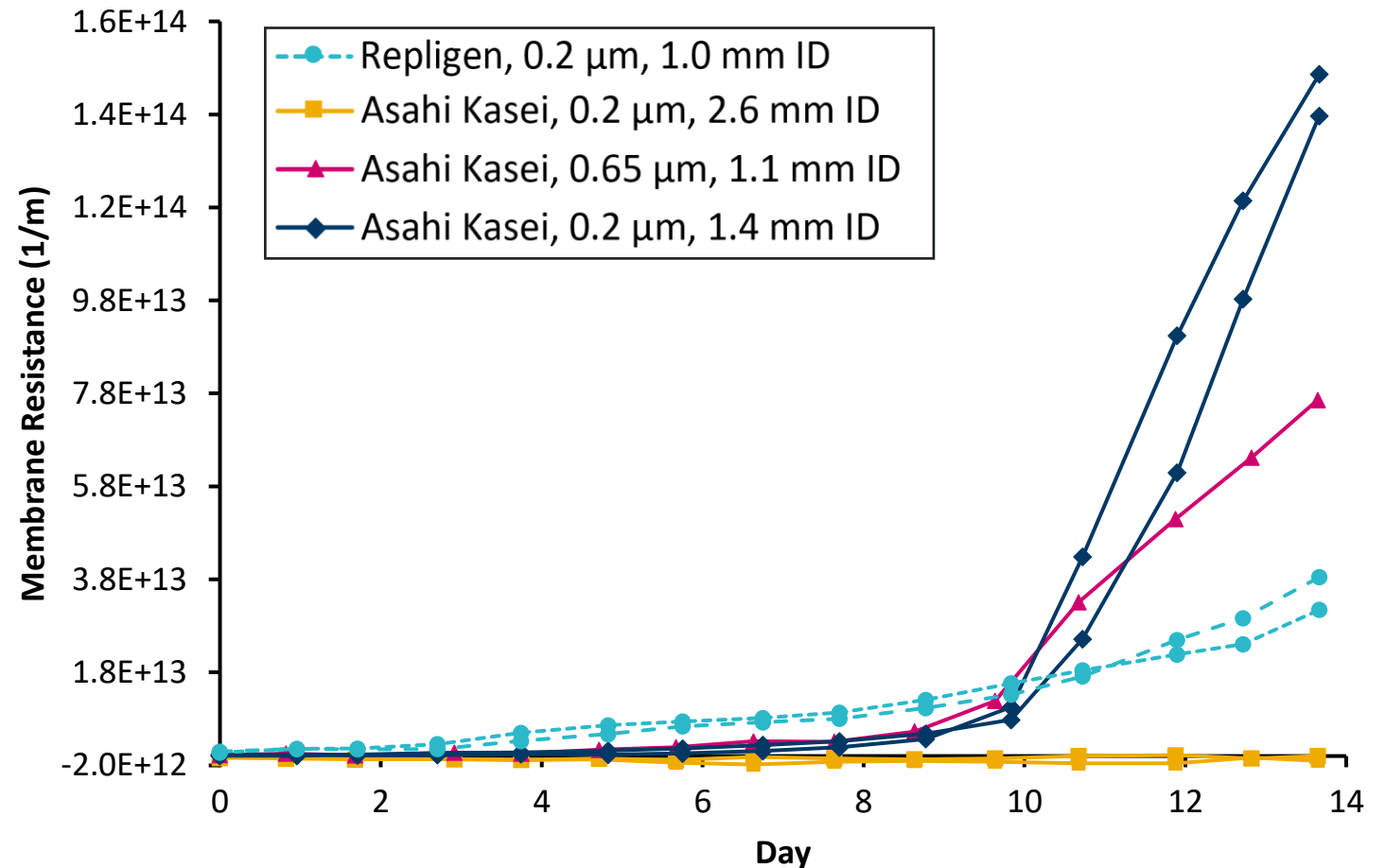
$$R_t = \frac{8\delta_m}{r_p^2 \varepsilon}$$

$\delta_m$  = membrane thickness

$\varepsilon$  = membrane porosity

$\mu$  = viscosity

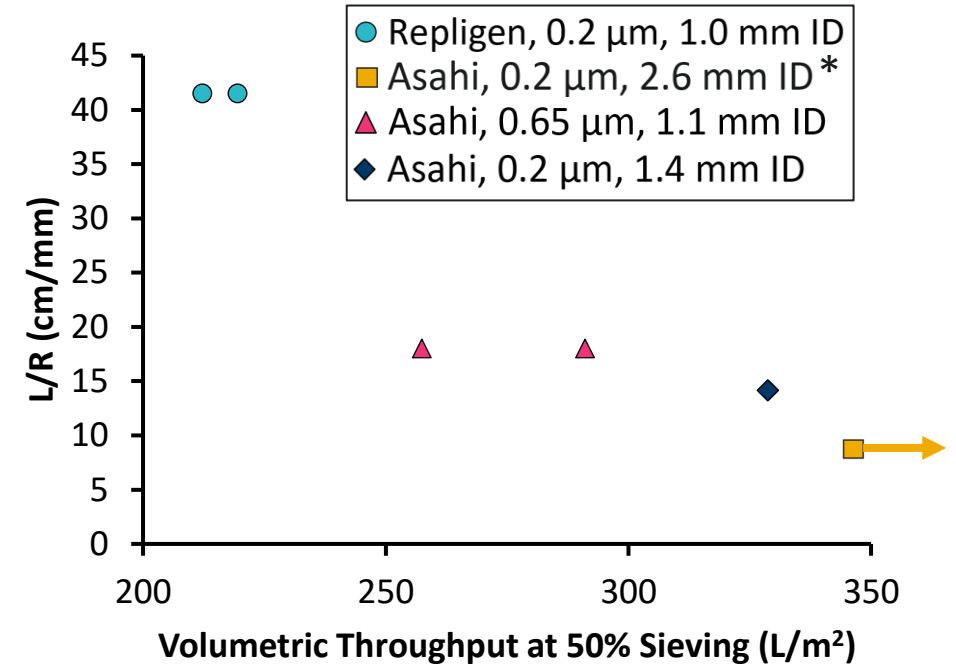
$r_p$  = pore radius



## Reduced flux: conclusions

- UMP-1047R (0.2  $\mu\text{m}$ ) filter maintained higher sieving than the UJP-1047R (0.65  $\mu\text{m}$ ) filter
  - Previous literature<sup>1</sup> suggests 0.2  $\mu\text{m}$  filter has minimal fouling due to wide-surface conical shaped pores measured by SEM that are not seen in 0.65  $\mu\text{m}$  filter
- UMP-153 (0.2  $\mu\text{m}$ ) filter maintained the highest product sieving
  - Largest lumen ID (2.6 mm) combined with operating at half the filter shear rate further reduces axial pressure drop and starving flow

*\*maintained 100% sieving at 350 L/m<sup>2</sup> throughput*



$$\gamma \propto Q \Rightarrow \Delta P \propto Q$$



# Overall summary

- Optimized filter geometry combined with operating at a lower shear rate improves filter performance and scalability

Larger lumen ID, shorter fibers, lower shear rate

Improves product sieving

Lowers hydraulic membrane resistance

Higher volumetric throughput per sieving %

Larger pore size

Lowers hydraulic membrane resistance



# Acknowledgements

- **BTE-Upstream**

- Ken Lee
- Dominique WuDunn
- Jimmy Vu

- **BTE-Modeling**

- Ashna Dhingra

- **Support functions**

- Media prep
- Bioreactor prep

- **Sponsors**

- Jon Coffman
- Venkat Raghavan

